Ulit-BiDet: An Ultralightweight Object Detector for SAR Images Based on Binary Neural Networks

Han Pu, Zhengwen Zhu, Qi Hu, and Dong Wang, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) target detection has extensively utilized convolutional neural networks (CNNs). Nonetheless, CNN-based methods often achieve favorable detection accuracy at the cost of high model complexity, hindering the deployment of the algorithm in real-time application scenarios, such as maritime rescue and military decision-making. To deal with this problem, we are the first to propose an ultralightweight object detector named Ulit-BiDet for SAR images, which incorporates a binary neural network with very low storage and computation costs. First, we creatively design a performance and cost-scalable binary backbone that adapts to the diverse resources and computational capacities of practical devices. Then, the backbone structure is optimized with a new nonlocal module to enhance semantic contextual information, thereby alleviating false detections caused by the interference from land clutter and sea clutter. Third, considering the SAR imaging mechanism, the interference near the ship boundary with similar scattering power probably affects the localization accuracy due to the interfered object-related contour information. To tackle the localization issue, we uniquely propose to utilize valuable and extra object-related contour semantics to guide representation learning of ship targets. The scheme compels the model to generate features that highlight object contour, thereby promoting accurate boundary localization in ship target detection. We validated the robustness of the proposed network in three mostly cited publicly available datasets. Experimental results demonstrate that our model achieves 97.2%, 95.2%, and 77.3% detection accuracy with only 1.27M parameters and 0.22G operations (Ops) on SSDD, SAR-Ship, and AIR–SARShip2.0 ship detection datasets, respectively.**

***Index Terms*— Binarized quantization, ship detection, synthetic aperture radar (SAR), ultralightweight network.**

I. Introduction

SYNTHETIC aperture radar (SAR) possesses the crucial attributes of providing all-weather and all-time imaging capabilities. Consequently, SAR has attained significant global recognition and has emerged as an indispensable tool for Earth observation, such as land cover classification [1], [2], [3] and object detection [4], [5], [6]. Among these applications, ship target detection for the SAR data is fundamental to many various essential scenarios, such as the identification of smuggling and illegal fishing vessels, monitoring and management of maritime transport, and the improvement of early warning systems for coastal defense [7], [8], [9], [10]. Existing research utilization of SAR for ship detection can be divided into two main categories: the first category involves object detection of SAR data after imaging [11], [12], while the second category involves target sensing directly on raw SAR data before imaging [13], [14]. Our research primarily focuses on the first category, specifically conducting lightweight detection studies. Recent advancements in marine target detection and surveillance employing SAR images have gained considerable focus within the field of remote sensing. During the initial stages, SAR-based ship detectors were designed with handcrafted features using methods such as constant false positive rate [15], [16], [17] and maximum stable region method [18] after sealand segmentation. The performance of these conventional methods is significantly affected by the statistical modeling of sea clutter and the estimation of parameters for the chosen model. However, as the model complexity increases and migration capability diminishes, the computational process becomes arduous. Recently, convolutional neural networks (CNNs) have been applied to the detection algorithm to improve detection accuracy in the SAR image ship detection field.

Generally, existing studies in the field of SAR image detection focus on solving three categories of problems: 1) the characteristics of multiscale and large-scale variation of SAR ship targets [11], [12]; 2) the detectability of small targets [19], [20]; and 3) the interference of complex land background [21], [22]. However, current methods all use deep CNNs for feature extraction, which results in a complex network structure and significant computational demands. Regrettably, none of them considers the constraints imposed by limited resources for achieving real-time detection, a critical requirement in numerous practical scenarios, including maritime rescue and emergency military decision-making.

Recently, many studies were dedicated to designing ultralightweight networks for object detection in optical images [23], [24], [25], [26]. However, there are only a few approaches that have addressed the development of lightweight ship detectors for SAR images [27], [28], [29]. Unfortunately, these methods have not truly achieved a proper balance between the accuracy and lightness of the model. Furthermore, the level of lightweightness achieved by these approaches remains inadequate to meet the resource limitations of spaceborne or airborne onboard embedded computing devices.

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The authors are with the Institute of Information Science and the Beijing Key Laboratory of Advanced Information Science and Network Technology, Beijing Jiaotong University, Beijing 100044, China (e-mail: ipuhan@bjtu.edu.cn; 22140608@bjtu.edu.cn; qihu_hbu@163.com; wangdong@bjtu.edu.cn).

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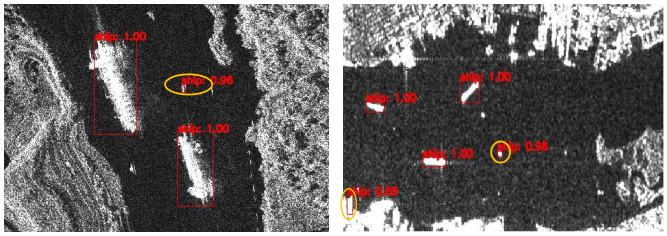


Fig. 1. Some detection results of the SOTA detectors of YOLOv4 [36] and Quant-Det [37], respectively. The red rectangles are the detection results. The yellow circles represent falsely detected bounding boxes, which is interference from land clutter and sea clutter.

Therefore, it is imperative to devise ultralightweight object detectors tailored specifically for SAR image ship targets, enabling seamless deployment on embedded platforms with limited resources.

Among all the lightweight neural network architectures, binary neural networks (BNNs) have emerged as a promising avenue. By imposing a strict bit-width reduction to only 1 bit, BNNs have successfully enabled real-time processing speeds on low-cost and energy-efficient embedded devices [30], [31], [32], facilitating the creation of binary object detectors [33], [34], [35]. These advancements, however, have been primarily focused on optical images and have not taken into account the distinctive characteristics of SAR images. Consequently, there is a need to address the design of an ultralightweight object detector specifically tailored for SAR images, leveraging the capabilities of BNNs. Designing such a detector presents certain challenges, particularly when considering the SAR imaging mechanism. Notably, one key problem is that target ship objects are often sparse and vulnerable to interference from land clutter and sea clutter in large scenes, which usually causes different degrees of false detection (yellow circles), as shown in Fig. 1. Conventional detection algorithms typically prioritize local ship features while overlooking the contextual relationships within the SAR image. Consequently, effectively extracting contextual semantic features from the SAR image becomes a formidable task. Another key problem is that the semantic representations of the local areas close to the ship contours are vulnerable to nearby interference with similar scattering power. As a result, the presence of interference disrupts the object-related contour information, which is crucial for the accurate location of certain target ships.

To solve the above problems, this article proposes an ultralightweight object detector for SAR images based on BNNs, namely, Ulit-BiDet. The proposed method offers scalability in terms of performance and cost, accommodating a wide range of resources and computational power available on practical devices. To the best of our knowledge, this study is the first attempt to incorporate binary quantization to design an ultralightweight SAR ship detector. Through extensive experimentation on SAR ship detection datasets, namely, SSDD [38], AIR-SARShip2.0 [39], and SAR-Ship [40], our approach shows comparable performance to existing methods while substantially reducing the number of parameters and computational efforts, demonstrating the effectiveness and ultralightweight of our model. The main contributions of this work are summarized as follows.

1) We design a performance and cost-scalable binary backbone, where the depth and width of the designed backbone are scalable to adapt to practical engineering applications with various resources and computational powers.
2) To address the problem of interference of land clutter and sea clutter in SAR images, the designed binary backbone incorporates nonlocal modules for semantic contextual feature extraction effectively, which alleviates false detection.
3) Considering that the semantic representations of the local areas close to the ship boundary are susceptible to nearby interference with similar scattering power, we carefully devise a contour-information enhancement module. This module enhances object contour semantics and explores valuable and powerful feature representations for SAR images, which can benefit ship target localization.
4) To promote the consistency between classification and location, a relation-constrained loss is designed to constrain the relationship between the classification confidence score and the localization qualities, which can effectively filter any biased predictions.
5) We conduct extensive experiments to verify the effectiveness of Ulit-BiDet and scale the model size to create tiny/small/medium/large models for various application scenarios. Ulit-BiDet achieves a better parameter–accuracy tradeoff than previous methods, demonstrating superior performance on the SSDD dataset. Specifically, our method with a large model (referred to as Ulit-BiDet-L) achieves detection accuracies of up to 97.2% on the SSDD dataset while only using 60× lower parameters and 80× lower OPs than the state-of-the-art (SOTA) lightweight ship target detector, Quant-Det [37]. Besides, our tiny model (referred to as Ulit-BiDet-T) only has a 0.8% drop in mAP while using only approximately one-sixth and one-fifth of the number of parameters and OPs over the lightweight network GhostNet [41]. In addition, our Ulit-BiDet-L delivers the mAP scores of 95.2% and 77.3% on the SAR-Ship and AIR-SARShip2.0 datasets, respectively, with magnitude reduced resource cost compared to the latest work of Quant-Det [37].

The remaining parts of this article are arranged as follows. Section II introduces the related works. Section III presents the Ulit-BiDet method. Section IV gives the experimental results on three datasets and the relevant analysis. Finally, conclusions are made in Section V.

II. Related Work

A. Ship Target Detection

Traditional methods for SAR ship detection typically employ CFAR algorithms [42], designed to adaptively adjust detection thresholds based on predetermined false alarm rates and the statistical properties of background clutter. Later, a notable advancement was made by Wang and Liu [43], who enhanced CFAR by incorporating both intensity and spatial information. Despite these strides, existing algorithms grapple

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with complex backgrounds inherent to SAR images. For example, ship targets may be in a complex ocean background and dock in a complex land background. As a result, these algorithms often struggle to handle such scenes effectively, and complex background clutter statistics frequently hamper the algorithm's performance. To tackle this issue, Huo et al. [18] selected candidate detection regions by calculating exceptionally stable extreme value regions and obtained thresholds to detect ships. Likewise, Shi et al. [44] separated ships from the background by calculating the directed gradient histogram, which led to better feature extraction. Nonetheless, these traditional approaches require manual parameter setting and adjustment and heavily rely on manual design. As a result, these traditional algorithms lack robustness for application across various SAR datasets [38], [39], [40]. In addition, their limited detection speed and real-time performance do not meet the demands of engineering applications.

The advantages of employing deep learning-based algorithms for end-to-end target detection in SAR images are becoming increasingly apparent. Unlike traditional algorithms that require manual feature extraction, deep learning methods streamline the training and detection process. They only focus on designing a robust network structure to extract the features of SAR targets effectively. Thus, CNNs have gained considerable attention for handling SAR image targets, particularly for identifying ships in SAR images. For example, to tackle the multiscale problem of SAR image ships, Liu et al. [12] created a ship proposal box generator to enhance both the recall rate and the quality of the proposal boxes. To improve detection and reduce misdetection of small targets in SAR ship images, Kang et al. [20] applied a multilayer fusion-based contextual region CNN in SAR ship detection. To alleviate the interference of complex land background, Yang et al. [22] introduced and embedded the coordinate attention module into the backbone. This embedment enhances the model to obtain stronger semantic features that better represent ship target objects. Nevertheless, these detection methods often rely on a very deep network model structure without considering practical engineering applications. As a result, the tradeoff between the number of parameters, floating-point operations (FLOPs), and detection accuracy remains inadequately balanced.

B. Lightweight Ship Target Detection

Recently, lightweight ship detection networks have gradually started to draw greater attention. Li et al. [28] improved R-CNN by incorporating both the inception module and skip connection, aiming to construct a lightweight basis network. Wang et al. [45] applied a lightweight network-based segmentation algorithm for image postprocessing and incrementally reduced false alarms via multilevel feature cascading. Zhou and Piao [46] substituted the backbone network of YOLOv4 with a lightweight network and used separable convolution to replace the standard one, which can effectively achieve higher accuracy with fewer parameters. However, the parameters still cannot meet the practical needs. Chang et al. [47] adopted YOLOv2 [48] to create an SAR ship detector with fewer parameters. While the detection speed is increased, the theoretical explanation is unclear and its performance is not very good. Zhou et al. [49] developed a lightweight meta-learning method for few-shot object detection on SAR images based on YOLOv3 [50]. Recently, Wan et al. [29] introduced a lightweight anchor-free method for SAR ship detection that reduced the model's parameters. Long et al. [27] proposed a lightweight model Lira-YOLO for ship detection in radar images, which uses the idea of dense connection and group convolution. Furthermore, Quant-Det [37] proposes an efficient and lightweight target detection network incorporating 8-bit soft quantization. Although these methods have contributed to the design of lightweight SAR ship detectors, they have not truly achieved a proper balance between the accuracy and lightness of the model. Moreover, they do not consider the need to design performance and cost-scalable detection models for real engineering applications with various computational resources.

C. Binary Neural Network

In extreme scenarios, binarization of the parameter and activation can yield up to $32\times$ and $64\times$ reductions in storage cost and computational cost [51], respectively. Consequently, BNNs have lately received great attention for deployment on constrained-resource devices [31], [52], [53]. In pioneering studies of BNNs, the binary weight network (BWN) [54] utilized binarized weight representations of $+1$ and -1, and the XNOR-Net [51] network applied binarization on both weights and activations (W/A). Nonetheless, they encounter the problem that training does not converge easily and results in accuracy degradation. To mitigate accuracy degradation, the recent work of BiRealNet [55] considered the mismatch between the gradient of the binarization function (e.g., sign) and STE (e.g., clip) in the backward propagation, by introducing an ApproxSign function to alleviate the gradient error. Meanwhile, the authors also incorporated a simple shortcut into the neural network to retain the information by leveraging a small number of real-valued activations, which could enhance the representational capacity of the binary quantized neural network substantially. Based on the observation that the performance of binary networks is sensitive to activation distribution variations, the study of ReActNet [56] proposed an extension to the sign function and the PRelu layer, referred to as the RSign function and the RPRelu layer, respectively, to facilitate effective learning of the distribution reshape and shift at near-zero extra cost. According to the above introduction, it is necessary and valuable to devise an ultralightweight SAR ship object detector incorporating binary quantization.

Therefore, considering the demands of real engineering applications, we innovatively design an ultralightweight algorithm for ship detection in SAR image datasets based on BNNs. The performance and cost of the algorithm are scalable to transplant to practical engineering applications that involve diverse resources and computational powers.

III. Method

A. Overview of the Method

Fig. 2 illustrates the structure of the proposed detector framework of Ulit-BiDet for SARs detection based on BNNs. The framework consists of five neural network parts: the

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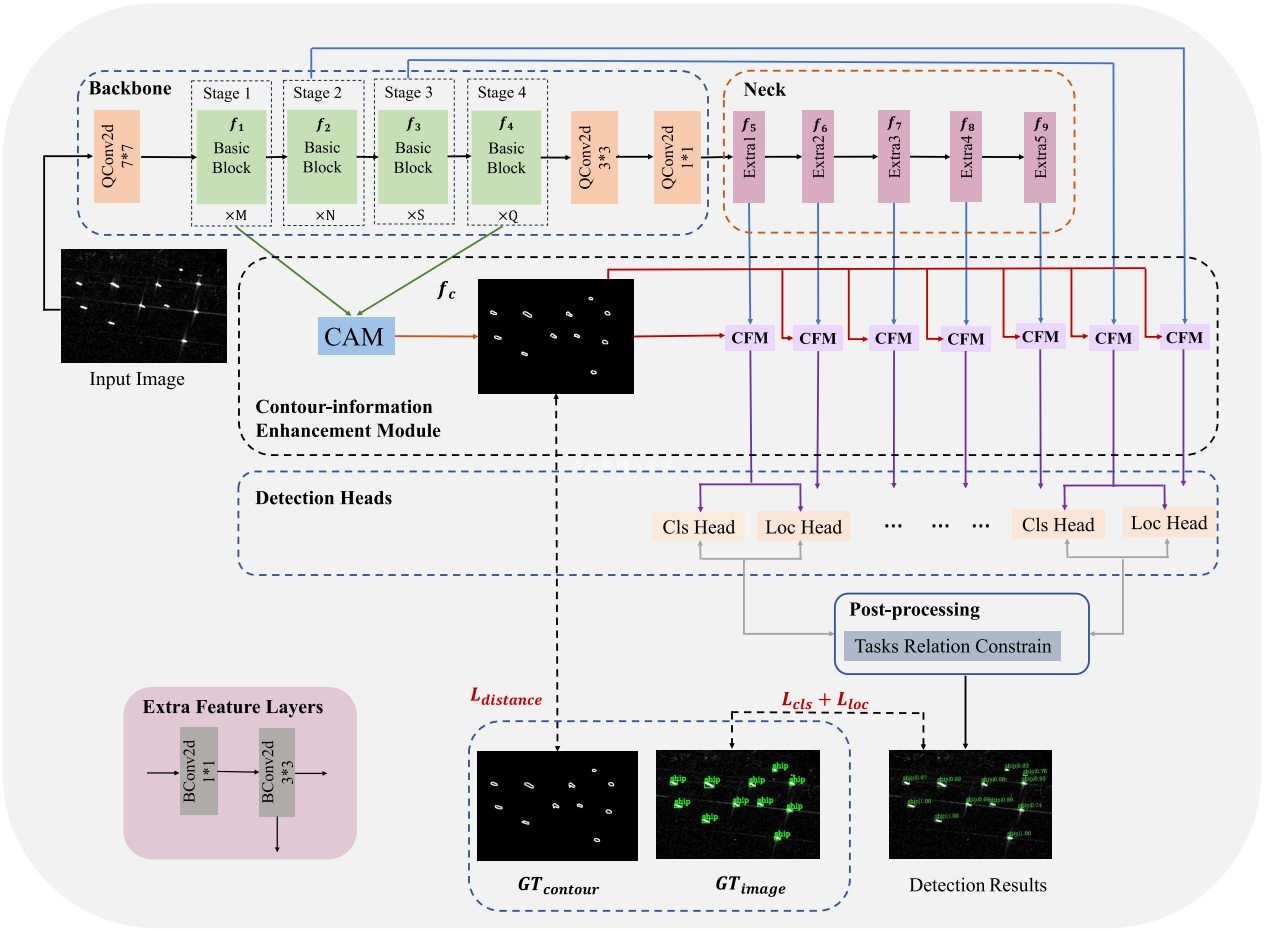


Fig. 2. Illustration of the network architecture of Ulit-BiDet, which consists of five key components: the built scalable backbone, feature-enhanced neck, the contour-information enhancement module, detection heads, and the postprocessing module. The backbone is built of four basic blocks, which are labeled with "M," "N," "S," and "Q." BConv2d represents the binary quantization, while QConv2d represents the 4- or 8-bit quantization.

performance and cost-scalable backbone, the feature-enhanced neck, the contour-information enhancement module, the detection heads, and the postprocessing module, respectively. First, the role of the built scalable backbone network is to extract multiscale feature maps with contextual semantic information from the various input images. Second, the extracted feature maps are fed into the neck part for feature fusion and enhancement. Then, the contour-information enhancement module is specifically devised to explore valuable and extra object-related contour semantics to guide representation learning of ship targets. By doing so, the module encourages the generation of features that highlight object structure, thereby promoting ship target detection of accurate contour localization. Following that, the generated enhancement feature maps are fed into the detection heads for object classification and location prediction. Finally, we design a relation-constrained loss to effectively constrain the relation between the classification task and location task, which benefits the subsequent nonmaximum suppression (NMS) postprocessing required to eliminate the redundant detected bounding boxes. We introduce the improvements of these five aspects in more detail as follows.

B. Performance and Cost-Scalable BNN Backbone

To adapt to practical engineering applications with various resources and computational powers, we design a performance and cost-scalable binary backbone, where the depth and width of the designed backbone are scalable. Besides, to address the problem of interference of land clutter and sea clutter in SAR images, the designed binary backbone incorporates nonlocal modules for semantic contextual feature extraction effectively, which alleviates false detection. In Fig. 3, we visualize the loss landscape of the BNNs and provide our observations and insights. We then present the two core modules of the binary backbone architecture design and the scalable structure construction of the backbone.

1) Optimized BNNs Architecture: For a full-precision CNN, the basic operation (OP) is given as follows:

$$A_r^l = W_r^l * A_r^{l-1} \tag{1}$$

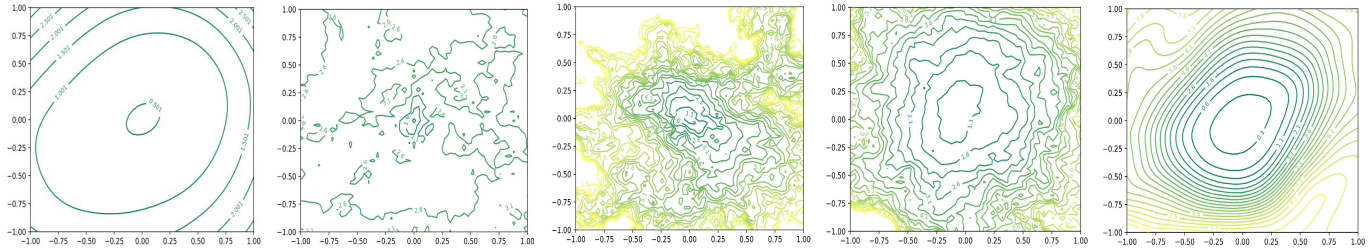
where W_r^l and A_r^{l-1} denote the real-valued weights of the lth layer and activations of the $(l-1)$th layer. r means the real value, and * represents the convolution OP. In BNNs, W_r^l and A_r^l are quantized into W_b^l and A_b^l using a specially designed sign function

$$W_b^l = \text{Sign}(W_r^l) \tag{2}$$

$$A_b^l = \text{Sign}(W_b^l \odot A_b^{l-1}) \tag{3}$$

where $W_b^l \in \{-1, +1\}$ and $A_b^l \in \{-1, +1\}$. b denotes binary value, and the costly convolution OPs * can be substituted by simple bit-count and XNOR OPs $\odot$, as explained in [51].

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(a) (b) (c) (d) (e)

Fig. 3. Loss landscape of BNNs and its full-precision counterpart ResNet-18 (contour line view) on the CIFAR10 dataset. We follow the setting of [57]. "scaling factor" and "nonlocal module" are the module designs in the proposed scalable backbone, while baseline denotes the ordinary binary network ReActNet-BiR18 [56] without optimization by both module designs. (a) ResNet-18. (b) Baseline. (c) Baseline + scaling factor (ours). (d) Baseline + nonlocal module (ours). (e) Ours.

To overcome the nondifferentiability of the standard sign function, a new differentiable function, namely, ApproxSign, was devised by BiRealNet [55] to calculate the approximate gradients as defined in the following:

$$\text{ApproxSign}(A_r^l)=\begin{cases}-1, & \text{if } A_r^l<-1\\ 2A_r^l+(A_r^l)^2, & \text{if } -1\le A_r^l<0\\ 2A_r^l-(A_r^l)^2, & \text{if } 0\le A_r^l<1\\ 1, & \text{otherwise.}\end{cases} \tag{4}$$

As defined by (5), the gradients can then be precisely calculated and applied to update the real-valued weights in the backpropagation stage

$$\frac{\partial\text{ApproxSign}(A_r^l)}{\partial A_r^l}=\begin{cases}2+2A_r^l, & \text{if } -1\le A_r^l<0\\ 2-2A_r^l, & \text{if } 0\le A_r^l<1\\ 0, & \text{otherwise.}\end{cases} \tag{5}$$

Observing that minor variations in activation distributions can have a big impact on the representation of semantic features in 1-bit CNNs, ReActNet [56] substitutes the conventional sign function and PReLU with RSign and the RPReLU functions. RSign was defined as a sign function with channelwisely learnable thresholds α_l

$$A_b^l=h\left(A_r^l\right)=\begin{cases}+1, & \text{if } A_r^l>\alpha_l\\ -1, & \text{if } A_r^l\le\alpha_l\end{cases} \tag{6}$$

where A_r^l denotes the real-valued input of the RSign function h in the lth channel, A_b^l denotes the binary output, and α_l is a learnable coefficient controlling the threshold.

Similarly, RPReLU is defined as

$$f\left(A_r^l\right)=\begin{cases}A_r^l-\gamma_l+\zeta_l, & \text{if } A_r^l>\gamma_l\\ \beta_l\left(A_r^l-\gamma_l\right)+\zeta_l, & \text{if } A_r^l\le\gamma_l\end{cases} \tag{7}$$

where γ_l and ζ_l are learnable shifts for adjusting the distribution of A_r^l, while β_l is another learnable coefficient that controls the slope of the negative part. All the coefficients are expected to be different across channels.

In summary, the aforementioned designs [55], [56] have attempted to solve the two main bottlenecks of BNNs: capacity degradation and optimization difficulty. However, the underlying reason behind these optimization bottlenecks of BNNs remains unclear.

It was pointed out in [58] and [59] that sharp local minima in the training loss landscape could negatively affect the generalization of deep networks, by which they will be more susceptible to shifts between train and test losses. Therefore, later, Liu et al. [56] explored how training strategies and optimizers affect BNNs by visualizing the loss landscape. Inspired by this idea, we will also use it as an indicator to guide the design of the backbone BNN. In this work, we select ReActNet-BiR18, which applied the ReAct [56] OPs to BiRealNet-18 [55], as the starter point (baseline architecture) of the backbone design process. To assess the characteristics of each optimized network architecture, we plotted the corresponding 2-D loss landscapes (contour line view), as shown in Fig. 3. By comparing Fig. 3(a) and (b), we can see that directly binarizing the network [56] makes the loss landscape surface extremely discontinuous and rugged; the contour lines are unstructured and are unaligned. The highly rugged landscape surface makes BNNs easier to converge into suboptimal minima and more sensitive to inappropriate optimization processes. Therefore, as seen from the loss landscape, the baseline network is optimized by the following schemes.

2) Learnable Scaling Factor: To facilitate the information propagation and increase the ability to capture complex patterns in SAR images, we construct a new binary convolution module that incorporates the learnable and adaptive scaling factor to conduct information coupling, as shown in the left part of Fig. 4. During the forward pass, the output distribution of a binary convolution is adjusted with a subsequent BatchNorm layer. Meanwhile, we first explicitly calibrate the information flowing through the learnable scale factor denoted as SF (red font view). This scale factor is equipped with a shortcut connection that connects the real activations (after the 1-bit convolution and/or BatchNorm layer, before the next 1-bit convolution layer) to activations of the consecutive block.

Compared to the baseline, our design offers a significantly more flexible shortcut connection mechanism, which dynamically learns to fuse features effectively to compensate for the limited representational capacity caused by binary quantization in SAR images. We observe that the loss landscape also becomes more uniform and smooth [see Fig. 3(c)], indicating that the scaling factor can effectively facilitate optimization.

3) Nonlocal Module: To alleviate the interference of land clutter and sea clutter in SAR images, we focus on improving the backbone's ability to extract contextual features. Motivated

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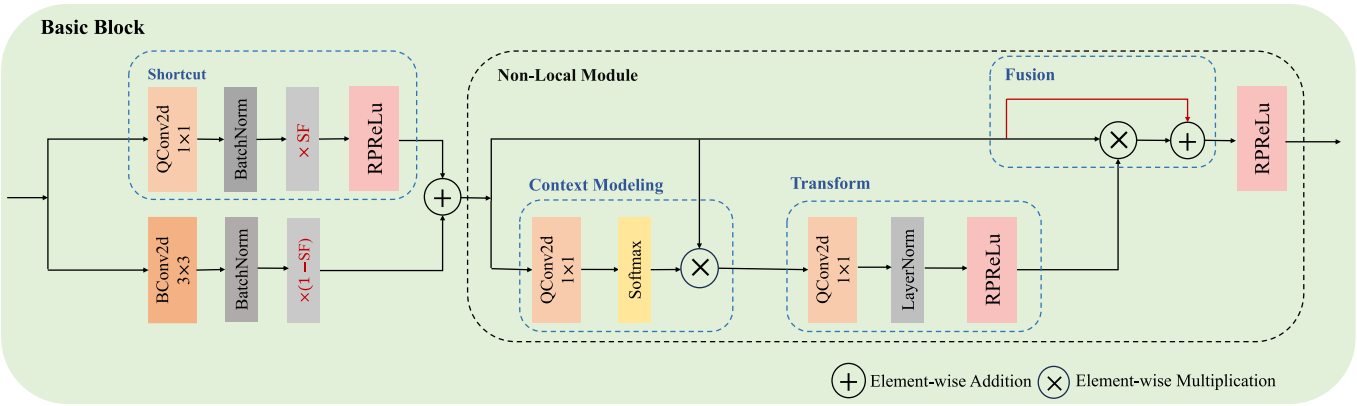


Fig. 4. Designed basic block that includes a learnable and adaptive scaling factor, denoted as scale factor SF, to fuse the consecutive block information dynamically and a new nonlocal module to model the contextual information. QConv2d represents the 4-bit quantization, and SF denotes the scaling factor.

TABLE I

CORRESPONDING PARAMETER OF THE DIFFERENT STAGES WITH DIFFERENT BASE WIDTHS IN BACKBONE ARCHITECTURE. THE PARAMETER OF STAGE M WITH BASE WIDTH 32 IS DENOTED AS 1.0P, AND THE RELATIVE CORRELATION OF PARAMETER NUMBERS BETWEEN DIFFERENT STAGES WITH VARIOUS BASE WIDTHS IS HIGHLIGHTED IN BLUE

Stage \ Base Width	32	48	64
M	0.49K(1.0P)	1.00K(2.0P)	1.68K(3.4P)
N	1.68K(3.4P)	3.63K(7.4P)	6.18K(12.6P)
S	6.18K(12.6P)	13.31K(27.2P)	23.12K(47.2P)
Q	23.63K(48.2P)	50.80K(103.6P)	89.24K(182.1P)

by GCNet [60], we propose a new nonlocal module to model the long-range dependencies to extract contextual information, which is illustrated in the right part of in Fig. 4. Specifically, the nonlocal module consists of three procedures: 1) context modeling, which denotes a 1×1 quantized convolution and is followed by the softmax function to obtain the attention weights; 2) feature transform via a 1×1 quantized convolution; and 3) feature fusion, which employs multiplication to aggregate the global context features to the feature of each position and then utilizes addition to aggregate the real activations through a skip connection (red line). Different from GCNet, to mitigate the computational burden, a single convolutional layer is employed in the architecture to conduct a transform function. Then, to compensate for the potential accuracy reduction resulting from the simplified transform, as well as to accommodate the characteristics of binary quantized networks, a skip connection (red line) is introduced to preserve the real activations and enhance the feature fusion. Specifically, the red line depicted in Fig. 4 connects the real activations (after the 1-bit convolution and/or BatchNorm layer, before the sign function) to activations of the consecutive block, through an identity shortcut. Consequently, compared to the standard 1-bit CNN without the identity shortcut, the representational capability of the nonlocal module with the feature fusion is significantly enhanced, and the additional cost of computation is negligible. As the quantized convolution increases the difficulty of optimization, we add layer normalization inside the quantized transform (before RPReLU) to ease optimization, as well as improve the generalization of the network, which can lead to better performance in complex scenes.

The designed nonlocal module can substantially improve the loss landscape, as shown in Fig. 3(d). Finally, by combining both the scaling factor and the nonlocal module, we obtain a significantly smoother loss landscape, as shown in Fig. 3(e), which is already very close to the full-precision backbone.

4) Scalable Structure Construction: To assess the performance and cost-scalable backbone architectures, the backbone family is constructed by stacking basic blocks under different stage design strategies. The overall architectures of the backbone family are illustrated in Fig. 2, while M, N, S, and Q denote the stage depth ratios. The number of parameters of a single stage can be calculated by $K_h \times K_w \times C_{\text{in}} \times C_{\text{out}}$, where K_h and K_w denote the height and width of the kernel size, respectively. C_{in} and C_{out} correspond to the input and output channels of the feature map. Table I illustrates the corresponding parameters (black font view) and relative parameters ratios (blue font view) of four stages (M:N:S:Q) across various base widths (32, 48, 64). It can be observed that the parameters exhibit significant variations with different stage design strategies.

Therefore, how to stack the basic blocks under different stage design strategies is still an open and valuable problem. Achieving a proper balance between the accuracy and lightness of the backbone is the key point of the problem. Therefore, we have thoroughly investigated the performance and resource cost of the backbone across varying stage depth ratios. As illustrated in Fig. 5, a collection of models with different stacked basic blocks has comparable levels of accuracy or parameter counts. Our selection rule encompasses two aspects: on one hand, selecting models with reduced parameter counts while maintaining similar levels of accuracy (i.e., model Backbone-T versus model E and model Backbone-L versus model G); on the other hand, prioritizing models that exhibit superior precision at similar parameter

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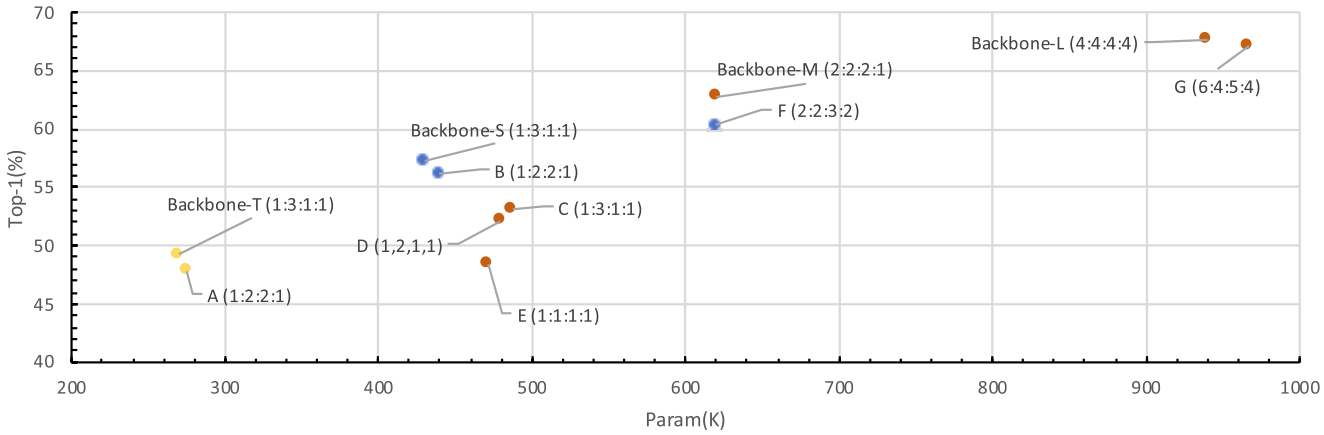


Fig. 5. Backbone performance of architectures with different stage depth ratios of (M:N:S:Q). The yellow, blue, and brown dots denote the backbone with base widths 32, 48, and 64, respectively.

TABLE II

BACKBONE ARCHITECTURE OVERVIEW AND CORRESPONDING PARAMS AND OPS AND PERFORMANCE ON IMAGENET. "QUA" REPRESENTS THE QUANTIZATION BIT WIDTH OF W/A IN FIG. 2 FOR THE FIRST LAYER, THE NONLOCAL MODULE, THE SHORTCUT BRANCH, AND THE OUTPUT LAYER, RESPECTIVELY

Models	Stage Ratios(M:N:S:Q)	Base Width(C)	Qua(W/A)	OPs (10^8)	Param(K)	Top-1(%)
Backbone-T	1:3:1:1	32	8/8-4/4-4/4-8/8	0.20	270	49.3
Backbone-S	1:3:1:1	48	8/8-4/4-4/4-8/8	0.31	430	57.3
Backbone-M	2:2:2:1	64	8/8-4/4-4/4-8/8	0.47	620	62.7
Backbone-L	4:4:4:4	64	8/8-4/4-4/4-8/8	0.65	940	67.6

levels (i.e., Backbone-S versus model B and Backbone-M versus model F). Specifically, as shown in Fig. 5, model E exhibits significantly higher parameters than the Backbone-T. However, the performance is still slightly lower than the Backbone-T. Similarly, model G achieves similar performance to the Backbone-L; however, the parameters of model G have an increase than the Backbone-L. Besides, model B has slightly higher parameters than Backbone-S, while the performance is lower by 1.10% compared to Backbone-S. Despite having similar parameters, model F exhibits a large reduced performance of 2.37% compared to Backbone-M.

In summary, according to the corresponding parameter cost and performance of the above models in Fig. 5, we choose the four stage ratios, as reported in Table II, to truly achieve a proper balance between the accuracy and lightness of the model. Table II shows the configuration of four backbone networks and corresponding top-one performance on ImageNet, where M, N, S, and Q denote the stage depth ratios, and C indicates the width of the input layer. The scalability design of the backbone can satisfy the real engineering application's specific needs and available resources.

C. Contour-Information Enhancement Module

Considering that the semantic representations of the local areas near the ship target contour, which are vulnerable to nearby interference with similar scattering power, the interfered object-related contour information probably impacts the localization accuracy. To tackle the localization issue, we uniquely propose the contour-information enhancement network to utilize valuable and extra object-related contour semantics to guide representation learning of ship targets. The contour-information enhancement network is built with two modules, including the contour-aware module (CAM) and the contour-guidance feature module (CFM). The CAM is used to explore edge semantics related to object contours, while the CFM is introduced to incorporate the explored contour features with the multiscale ship target features at various layers to guide the representation learning of binary detectors for SAR Images. Specifically, as illustrated in Fig. 2, CAM integrates the low-level feature (f_1) and the high-level feature (f_4) to model the object contours f_c (as shown by the green lines). Then, the CFM is designed to incorporate object contour cues f_c into the representation learning. As shown by the blue lines, all the multiscale feature maps, f_2, f_3, f_5, f_6, f_7, f_8, and f_9, are connected to the CFMs to enhance the feature representation with object contour semantics f_c. All the outputs of CFMs are connected to the detection heads to allow predictions at multiple scales.

1) Contour-Aware Module: Given the SAR imaging mechanism, the interference near the ship contours with similar scattering power often affects the localization accuracy due to the disturbed object-related contour information. For instance, the sea surface and the ship deck both have a relatively low scattering intensity, while some facilities in the harbors have similar strong scattering phenomena with the metal components of the ships. As a result, some predicted bounding boxes cannot encompass the complete object accurately [21]. We identify that the semantic representations of the local areas close to the ship contour are vulnerable to nearby interference with similar scattering power in SAR images, and the interference disturbs the effective learning of the object-related contour information. Consequently, the

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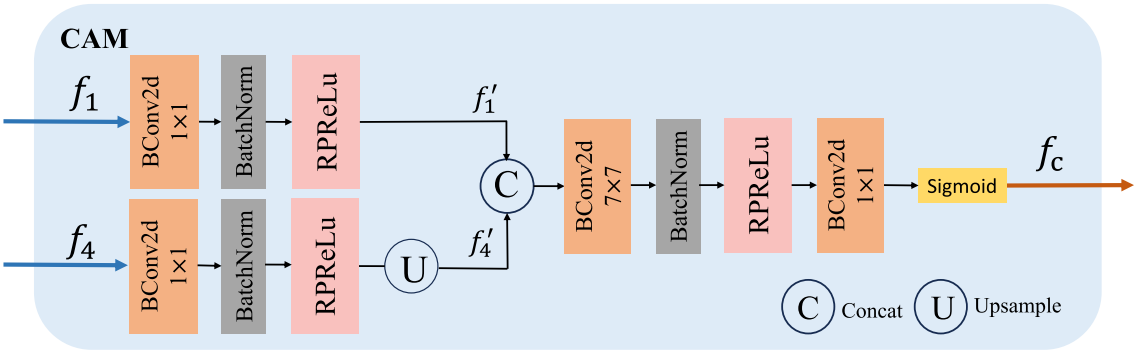


Fig. 6. Designed CAM.

disturbed object-related contour information results in an inaccurate bounding box.

Therefore, a good object contour prior is beneficial for object detection, especially in localization. Although low-level features contain extensive edge details, they also incorporate many nonobject edges. As a result, high-level semantic or location information is required to facilitate the exploration of object contour features. The fusion of low- and high-level features has been utilized in several applications, such as object detection [21], image segmentation [61], and image classification [62]. The fusion is to leverage the advantages of both types of features, i.e., the concrete details of low-level features and the abstract semantic information of high-level features, to obtain better performance. Inspired by this, we integrate the low-level feature (f_1) and high-level feature (f_4) to model the object contours, as shown in Fig. 2.

Especially, as depicted in Fig. 6, two 1×1 binary convolution layers are first applied to adjust the channels of f_1 and f_4 to 64 (f_1') and 256 (f_4'), respectively. Then, we incorporate the feature f_1' and the upsampled f_4' by concatenation OP. Finally, we obtain the contour feature f_c via a 7×7 binary convolution layer and one 1×1 binary convolution layer followed by the sigmoid function. The adoption of 7×7 binary convolution kernel captures a larger range of contextual information, allowing for the learning of more global features. Meanwhile, the 1×1 convolution kernel is mainly used to tune the number of channels in the network. Besides, the introduction of 1×1 convolutional layer followed by the sigmoid function can add nonlinearity without increasing the receptive field [63], which can improve the representation capacity of the network without significantly increasing the computational complexity.

2) Contour-Guidance Feature Module: The CFM is built to integrate boundary-related contour information into the representation learning to enrich the feature representation with object structure semantics. It is a well-known fact that semantics vary across different feature channels. Therefore, to achieve good integration and obtain an effective representation, we employ a local channel attention mechanism to investigate cross-channel interaction and mine the crucial cues between channels.

As depicted in Figs. 2 and 7, given the extracted multiscale features f_i ($i \in 2, 3, 5, 6, 7, 8, 9$) and the extracted contour feature f_c, we first take the elementwise multiplication between them with an additional skip-connection and a binarized 3×3 convolution to obtain the initial fused features f_i^c, which can be denoted as

$$f_i^c = \mathbf{F}_{\text{BConv}}((f_i \otimes D(f_c)) \oplus f_i) \tag{8}$$

where $\mathbf{F}_{\text{BConv}}$ is a binarized 3×3 convolution and D denotes the bicubic downsampling OP. $\otimes$ is elementwise multiplication, and $\oplus$ is elementwise addition.

To enhance feature representation, inspired by Wang et al. [64], we utilize local attention to explore essential feature channels. Specifically, we integrate the convolution features (f_i^c) by channelwise global average pooling (GAP). Then, we calculate the corresponding channel attention (weight) by the 1-D binarized convolutions followed by a sigmoid function. Unlike fully connected OPs, which capture dependencies across all channels but exhibit high complexity, we focus on a local cross-channel interaction and learn each attention in a local manner, e.g., only take into account its k adjacent channels of every channel. Following that, we multiply the channel attention with the input feature f_i^c followed by the BatchNorm and RPReLU [56] and then add the f_i^c to obtain the final output f_i^a, which can preserve the real activations before the sign function of the binary convolution to increase the representational capability of the 1-bit CNN, i.e.,

$$f_i^a = \sigma(\mathbf{F}_{1-\text{D}}^k(\text{GAP}(f_i^c))) \otimes f_i^c + f_i^c \tag{9}$$

where $\mathbf{F}_{1-\text{D}}^k$ is 1-D binarized convolution with kernel size k and σ denotes the sigmoid function. The kernel size k can be adjusted dynamically by $k = |(1 + \log_2(C))/2|_{\text{odd}}$, where C is the channels of f_i^c and $|*|_{\text{odd}}$ denotes the nearest odd number. The kernel size is proportional to the channel dimension. The proposed attention strategy can highlight critical channels and suppress redundant channels or noises, thereby enhancing semantic representation.

D. Relation-Constrained Loss

To promote the consistency between the classification and location of ship targets, a relation-constrained loss is designed to constrain the relationship between the classification confidence score and the localization qualities, which can effectively filter any biased predictions. In the postprocessing

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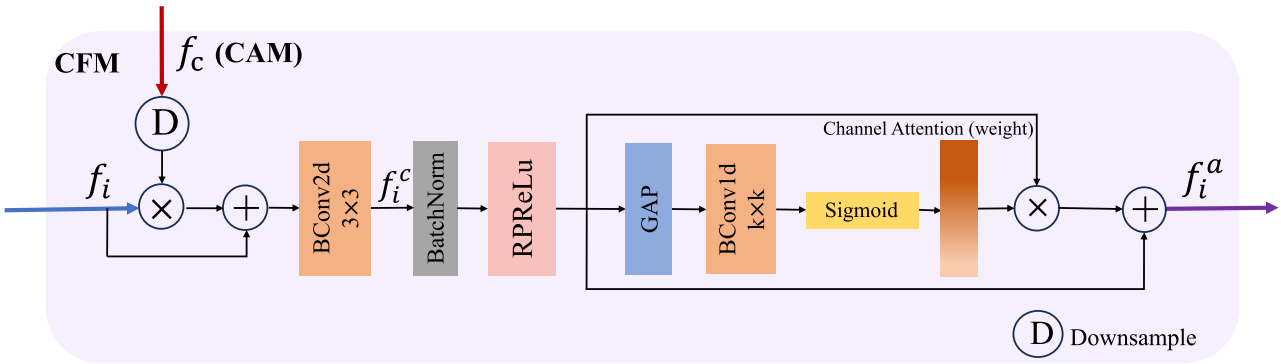


Fig. 7. Designed CFM. The box with color change is the obtained channel attention (weight).

stage, a majority of the detected boxes are redundant since the predefined anchors are densely located. Therefore, the NMS postprocessing [65] algorithm is required in most classical object detectors [48], [66], [67], [68], [69] to sort the detected bounding boxes according to their confidence scores and then eliminate all the redundant ones that have large Intersections over Union (IoUs) with the highly ranked box. As pointed out by IoU-Net [70], in optimal images, the measured small Pearson relation coefficient of the localization qualities (i.e., IoU scores) and its classification confidence score before NMS indicates that the localization accuracy is not positively correlated with the classification confidence. Thus, only using confidence score as the basis for bounding box selection in traditional NMS algorithm is not suitable, in which the basis predictions that are precise in classification but wrong in localization are likely to appear. Therefore, to avoid the flaws of the NMS, IoU-Net and ATSS [71] introduce an additional auxiliary head to predict IoU or centeredness. However, these optimized methods need to introduce additional learnable parameters.

In addition, unlike optical images, SAR exhibits a distinct imaging mechanism that often leads to significant scattering characteristics in ship targets within the images. Consequently, this phenomenon introduces substantial noise interference, which can adversely affect the correlation between the IoU and the classification confidence of the detected bounding boxes. For instance, noise interference can be falsely detected as a ship target, despite lacking corresponding ground-truth (GT) information. Consequently, establishing a more reasonable correlation between the classification scores and IoU of the detections within a single SAR image, prior to NMS, becomes crucial for enhancing the accuracy of the NMS algorithms in the detection flow.

To address this issue, we propose an alternative approach that focuses on directly regulating the correlation between the classification score and IoUs of the detected boxes, without introducing additional learnable parameters. Our proposed scheme leverages ordinal measurements, specifically the Kendall correlation [72], as the foundation for calculations aimed at enforcing correlation. We introduce the Kendall correlation loss, which serves as a straightforward plug-in loss term that is independent of the specific detector used. This loss term demonstrates the capability to enhance the performance of the object detectors proposed in this study. The construction of the proposed Kendall correlation loss term is given as follows:

$$L_{\text{relation}} = \frac{1}{|I|} \times \sum_{i \in I} \left\{ 1 - \text{Kendall}(p^i, \text{IoU}^i_{\text{detected boxes}}) \right\}. \quad (10)$$

Denoting the set of images to be evaluated by I, the newly introduced loss term aims to impose a constraint on the Kendall relationship between the confidence score p^i and the averaged IoU ($\text{IoU}^i_{\text{detected boxes}}$) across images. This constraint helps enhance the consistency of performance evaluation metrics employed in both the classification and location tasks, thereby mitigating biased selections within the NMS algorithm.

E. Model Objective

The proposed model incorporates two types of supervision: bounding box coordinate information GT_{image} and object contour information $\text{GT}_{\text{contour}}$, as illustrated in Fig. 2. The former GT_{image} is used to supervise the predicted bounding boxes, and the latter $\text{GT}_{\text{contour}}$ is utilized to supervise the extracted contour information f_c. Specifically, to generate the $\text{GT}_{\text{contour}}$, we adopt a two-step process. First, we employ the Canny operator to extract edge information from the SAR images. Subsequently, we apply a contour extraction function to the extracted information, deriving ship contours, denoted as $\text{GT}_{\text{contour}}$. Meanwhile, L_{distance} is utilized to measure the distance between $\text{GT}_{\text{contour}}$ and f_c, which is inspired by Xie et al. [73] to monitor the optimization of the model.

In total, besides the regular classification loss L_{cls} and the location loss L_{loc}, we add the proposed relation-constrained loss term L_{relation} for consistency supervision and the distance loss L_{distance} for contour supervision. Thus, the learning objective for Ulit-BiDet can be described by the following new loss function:

$$\begin{aligned} \min L_{\text{loss}}(t) = {} & \lambda_{\text{cls}} L_{\text{cls}}(\text{GT}_{\text{image}}, P) + \lambda_{\text{loc}} L_{\text{loc}}(\text{GT}_{\text{image}}, P) \\ & + \lambda_{\text{relation}} L_{\text{relation}}(p, \text{IOU}_{\text{detectionboxes}}) \\ & + \lambda_{\text{distance}} L_{\text{distance}}(\text{GT}_{\text{contour}}, f_c) \end{aligned} \quad (11)$$

where the hyperparameters λ_{cls} and λ_{loc} are used to determine the relative importance of the classification and location loss terms, respectively, within the overall loss function. Similarly,

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TABLE III
COMPARISON RESULTS FOR SAR SHIP DATASETS

	SSDD	AIR-SARship2.0	SAR-Ship
Number of images	1160	300	43819
Number of targets	2540	2040	59535
ANTI	2.2	6.8	1.36
Image size	about 500 × 500	1000 × 1000	256 × 256

ANTI represents the average number of objects in a single image

the hyperparameters $\lambda_{\text{relation}}$ and $\lambda_{\text{distance}}$ are employed to weigh the significance of the relation and distance loss terms, respectively, within the overall loss. In addition, P represents the predicted bounding boxes before the NMS postprocessing, while p and $\text{IOU}_{\text{detectionboxes}}$ correspond to the confidence score and IOU of the respective bounding boxes.

IV. EXPERIMENTS

A. Dataset

The experiments in this study were conducted using three different datasets: SSDD [38], high-resolution SAR ship detection dataset-2.0 (AIR-SARShip-2.0) [39], and SAR-Ship datasets [40]. These datasets encompass a wide range of scenes, including harbors, islands, and sea surfaces with varying levels of sea state. Table III provides detailed information regarding the comparison of these three datasets. For the SSDD dataset, which is the earliest publicly available dataset in terms of its release date and consists of 1160 images, we adopt the same scheme as Li et al. [38] to generate the training and test sets. The AIR-SARShip-2.0 dataset is an extension of version 1.0, featuring additional data. It comprises 300 images (size 1000 × 1000) obtained from the Gaofen-3 satellite. The SAR-Ship dataset consists of 43 819 images (size 256 × 256) acquired from both the Gaofen-3 and Sentinel-1 satellites. To maintain consistency with the division ratio of the SSDD dataset, we randomly select 80% of the entire dataset as the training set and allocate the remaining 20% as the test set for both the AIR-SARShip-2.0 and SAR-Ship datasets.

B. Implementation Details

1) Technical Details: To implement the proposed method, we utilized the PyTorch framework. The algorithm was executed within a deep learning environment established on an Ubuntu v16.04 system, specifically CUDA v11.1, cuDNN v8.0, and PyTorch 1.7.0. All experiments were conducted on Nvidia RTX 3090 GPUs with 24 GB of graphic memory. These GPUs provided the necessary computational capabilities for the efficient execution of the proposed method. The performance-scalable backbones were designed and initially trained on the ImageNet ILSVRC12 dataset [74]. Subsequently, the pretrained backbone network was imported for the proposed method (Ulit-BiDet), and further training was conducted for 300, 600, and 50 epochs on the SSDD [38], AIR-SARShip2.0 [39], and SAR-Ship [40] datasets, respectively. During training, the image size was set to 512 × 512 for the SSDD, AIR-SARShip2.0, and SAR-Ship datasets. This standardized image size allowed for consistent training and evaluation across the different datasets. During training, a batch size of 4 was utilized, and the Adam optimizer [75] was applied for all training processes. The initial learning rate was set to $1e-4$ and decayed using a polynomial learning rate scheduler, gradually decreasing it to 0 across all datasets. Regarding the hyperparameters, the values of λ_{cls}, λ_{loc}, $\lambda_{\text{relation}}$, and $\lambda_{\text{distance}}$ were set to 1, 1, 0.5, and 1, respectively. These values determined the relative importance of the classification loss, location loss, relation loss, and distance loss within the overall loss function.

2) Computational and Storage Costs: FLOPs was a widely adopted metric to represent the number of FLOPs involved in the real-valued convolutional layers, while QOPs [80] was often used to measure the number of fixed-point OPs in the quantized convolutional layers. BOPs specifically represent the binary OPs performed in 1-bit convolutional layers. Therefore, for the proposed model, the total OPs are calculated as OPs = BOPs/64 + QOPs (4-bit)/8 + QOPs (8-bit)/4 + FLOPs following Liu et al. [56]. This calculation accounts for the different types of OPs involved in the model architecture, providing a comprehensive measure of the total computational workload. The calculation of parameters (params) in this study also takes into account the different types of convolutional layers used. For real-valued convolutional layers, the number of parameters is counted directly. For 4- or 8-bit convolutional layers, the number of parameters is calculated by dividing the counted parameter numbers by 8 or 4, respectively. Similarly, for 1-bit convolutional layers, the number of parameters is calculated by dividing the counted parameter numbers by 32 as done in previous studies [55], [56]. The total number of parameters (total params) is then obtained by summing the parameter counts for the three types of convolutions mentioned above.

C. Ablations Study

To conduct a comprehensive analysis of our method's effectiveness, we have devised a four-part optimization strategy. This approach encompasses various components, namely, the optimized binary backbone network, CAM, CFM, and the relation-constrained loss. To assess the significance of each component, we performed an ablation experiment by comparing it to the baseline network Ulit-BiDet-B. We note that Ulit-BiDet-B is constructed by incorporating the direct binarized backbone, ReActNet-BiR18, along with a multi-scale feature extraction neck part and detection heads. The obtained results from this ablation experiment are presented and compared in Table IV. This comparative analysis allows us to evaluate the individual contributions of these components toward enhancing the overall performance of the detector.

1) Effectiveness of Scalable Backbone: The inclusion of the optimized binary backbone network resulted in a significant increase in mean Average Precision (mAP) by 1.6% (from 86.3% to 87.9%) compared to the baseline Ulit-BiDet-B. This improvement suggests that the dynamic learning of feature fusion effectively compensates for the limited representational capacity caused by binary quantization in SAR images. Furthermore, by enhancing the contextual information of features, the interference of land clutter and sea clutter in SAR images is reduced, leading to improved ship target classification and

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TABLE IV
EXPERIMENTAL RESULTS OF EACH PROPOSED COMPONENT ON THE SSDD DATASET. THE MODEL OF ULIT-BIDET-TINY WAS USED TO CONDUCT THE EXPERIMENTS

Model	Designed binary backbone	CAM	CFM	Realtion constrained loss	mAP(%)	Rec(%)	Pre(%)
Ulit-BiDet-B					86.3	87.7	69.8
Ulit-BiDet	✔				87.9	88.5	70.1
Ulit-BiDet		✔			88.2	84.9	72.8
Ulit-BiDet			✔		88.7	85.6	72.8
Ulit-BiDet		✔	✔		89.4	86.7	73.1
Ulit-BiDet	✔	✔	✔		90.5	88.7	77.6
Ulit-BiDet	✔	✔	✔	✔	91.3	92.2	76.9

TABLE V
ABLATION STUDY OF SCALING FACTOR AND NONLOCAL MODULE ON IMAGENET, SHOWING THE TOP-ONE ACCURACY

Method	w/o Non-Local Module	w/ Non-Local Module
w/o Scaling Factor	65.9	66.8
w/ Scaling Factor	66.3	67.6

TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT ATTENTION BLOCKS ON IMAGENET, SHOWING THE TOP-ONE ACCURACY. THE BASELINE DENOTES THE DIRECT BINARIZED NETWORK REACTNET-BIR18

Method	mAP(%)	Params(M)	OPs(10^8)
baseline	65.9	1.05	1.89
baseline+SENet [76]	66.0	1.14	1.97
baseline+ECANet [64]	66.4	1.05	1.97
baseline+CBAM [77]	66.2	1.14	1.97
baseline+External attention [78]	66.2	1.32	2.30
baseline+Self-attention [79]	66.6	2.21	3.22
baseline+GCBlock [60]	66.1	1.24	2.21
baseline+Non-Local Module (ours)	66.8	1.22	2.20

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT NORMALIZATION OPERATORS IN NONLOCAL MODULE

Activation	mAP(%)	Rec(%)	Pre(%)
BatchNorm	96.8	95.3	88.3
LayerNorm	97.2	97.6	88.9

TABLE VIII
EFFECT OF USING NONLINEARITIES FOR THE ACTIVATION OPERATOR IN CFM ON SSDD

Activation	mAP(%)	Rec(%)	Pre(%)
ReLU	96.7	97.1	88.6
Softmax	97.0	97.3	88.7
Sigmoid	97.2	97.6	88.9

reduced false detection. To further investigate the impact of the proposed learnable scaling factor and nonlocal module, we conducted another ablation study. The results, as presented in Table V, demonstrate the performance of various configurations. The baseline binary backbone, without either component, achieved an accuracy of 65.9% on ImageNet. Adding only the scaling factor to the baseline increased the accuracy by 0.4%. Similarly, incorporating only the nonlocal module led to an improvement of 0.9%. Finally, incorporating both designs resulted in the highest accuracy of 67.6%. This comparative analysis confirms the significance of the designed backbone, as both components effectively enhance the representational capacity of the model. These findings align with the expectations raised by the loss landscape comparison illustrated in Fig. 3.

In order to demonstrate the effectiveness and lightweight nature of our proposed nonlocal module, we conducted various experiments where we replaced the nonlocal module with the quantized SENet [76], ECANet [64], CBAM [77], external attention [78], self-attention [79], and GCBlock [60] on the ImageNet dataset. The results of this experiment are presented in Table VI. Through rigorous evaluation and analysis, we can observe that our nonlocal module exhibits 66.8% top-one accuracy, which is 0.8%, 0.6%, and 0.6% higher accuracy, respectively, even compared to similar parameter levels counterparts such as SENet, CBAM, and external attention. Besides, our designed module reduced almost 1M parameter counts while maintaining comparable levels of accuracy compared to self-attention. These results validate both the effectiveness and lightweight nature of our proposed nonlocal module in comparison to the various attentions.

In developing the proposed nonlocal module architecture, there were many alternative choices in network structure, such as the normalization operator in the feature transformation in nonlocal modules. To evaluate which type of normalization operator could deliver the best performance, we experimented to compare the performance of using batch normalization versus layer normalization for feature transformation in nonlocal modules, respectively, and the results are illustrated in Table VII. It is observed from the performance data that the layer normalization delivered better performance. This suggests that the selection of layer normalization in the proposed nonlocal module for feature transformation is appropriate.

2) Effectiveness of CAM: By incorporating the CAM module into our approach, we observed a significant increase in mAP score by 1.9% (from 86.3% to 88.2%) compared to the baseline Ulit-BiDet-B, as shown in Table IV. This improvement highlights the capability of the CAM module to enhance detection performance by leveraging contour

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TABLE IX
ABLATION STUDIES OF DIFFERENT INPUTS OF CAM ON THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	SSDD			AIR-SARShip2.0			SAR-Ship		
	mAP	Rec	Pre	mAP	Rec	Pre	mAP	Rec	Pre
$f_3 + f_4$	90.4	88.2	72.6	67.5	68.9	60.0	83.5	84.6	68.7
$f_2 + f_4$	90.6	88.5	72.5	67.8	69.2	60.1	83.8	83.8	**69.2**
$f_1 + f_4$(final)	**91.3**	**92.2**	**76.9**	**68.1**	**69.8**	**60.5**	**84.2**	**85.7**	69.1

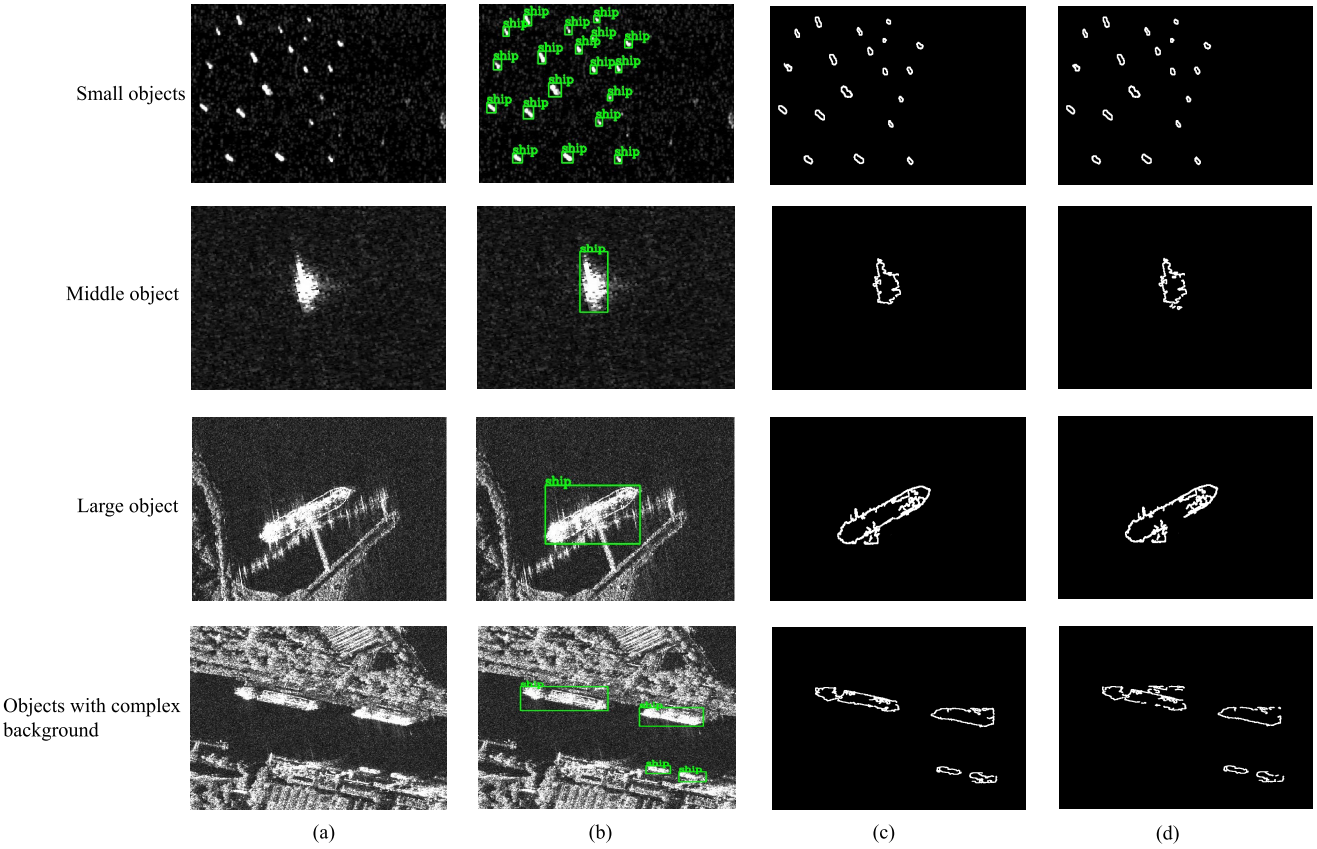


Fig. 8. Examples of object-related contour exploration in CAM. (a) Image. (b) GT_{image}. (c) $GT_{counter}$. (d) Contour information f_c.

information exploration. Specifically, the output f_c of the CAM is fused with multiscale features f_i (where i ranges from 2 to 9 except 4). The resulting fused features are then connected to the detection heads, enabling predictions at multiple scales. This integration of the CAM module allows for improved object detection accuracy and the ability to capture objects at different scales.

3) Effectiveness of CFM: The addition of CFM to the detector resulted in a notable improvement in mAP of 2.4% (from 86.3% to 88.7), as shown in Table IV, compared to the baseline Ulit-BiDet-B. The inclusion of CFM involves reducing the initial fusion OP and retaining only the local channel attention within CFMs. Subsequently, we incorporated the CAM module to validate the effectiveness of contour information integration. This integration further improved the model performance by 0.7% (from 88.7% to 89.4%). These results clearly demonstrate the effectiveness of the proposed CFM in enhancing the final predictions, as well as the impact of integrating contour information through the CAM module.

In the CFM block, the 1-D binarized convolution layer is followed by a sigmoid activation to ensure that the value of the learned channel attention (weight) is within the range of [0, 1]; thus, the choice of the activation function also has a conclusive impact on the learned essential feature channels. We next conducted experiments on setting different nonlinearity activation functions for the channel attention mechanism. Two other functions were considered: Softmax and ReLU, and the results are reported in Table VIII. It can be seen from the performance data that the sigmoid function delivered the best performance. This suggests that the selection of sigmoid in the proposed CFM architecture is appropriate. The unsatisfying performance of Softmax and ReLU may lie in that Softmax is well-suited for multiclass problems and ReLU's output range from 0 to infinity is not suitable for attention mechanisms.

4) Effectiveness of Relation-Constrained Loss: The integration of the relation-constrained loss in our approach resulted in an improvement in mAP by 0.8% (from 90.5% to 91.3%). This outcome demonstrates the effectiveness of incorporating the proposed loss in preserving the optimally

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TABLE X

PERFORMANCE COMPARISON WITH MODERN CNNS AND SOTA BINARY NETWORKS ON IMAGENET. "QOPS" DENOTE THE INT-4 AND INT-8 OPS IN TOTAL, WHILE "BOPS" REPRESENT THE BINARY OPS FOR THE 1-bit CONVOLUTIONAL LAYERS. MOREOVER, FLOPS REPRESENT THE NUMBER OF FLOPS INVOLVED IN THE REAL-VALUED CONVOLUTIONAL LAYERS

Binary Designs	W/A	BOPs(10^8)	QOPs (10^8)	FLOPs (10^8)	OPs (10^8)	Param (M)	Top-1(%)
ResNet18 [82]	32/32	-	-	46.92	46.92	11.13	70.7
ShuffleNetV1 [82]		-	-	2.95	2.95	2.6	68.6
XNOR-Net [51]	1/1	17.02	-	1.20	1.47	1.05	51.2
XNOR-Net++ [83]		17.02	-	1.20	1.47	1.05	57.1
BiRealNet-18 [55]		16.81	-	1.39	1.65	1.05	56.4
BiRealNet-34 [55]		35.33	-	1.39	1.94	1.28	62.2
ReActNet-BiR18 [56]		48.21	-	1.63	1.89	1.05	65.9
Backbone-T	1/1	1.30	0.64	-	0.20	0.27	49.3
Backbone-S		2.93	1.00	-	0.31	0.43	57.3
Backbone-M		6.36	1.38	-	0.47	0.62	62.7
Backbone-L		16.76	1.40	-	0.65	0.94	67.6

detected boxes for both classification and regression tasks simultaneously. To further validate the effectiveness of the proposed loss, we calculated the Kendall correlation coefficient (KCC) [72] between the confidence and IoU scores for our method and the baseline Ulit-BiDet-B. The KCC value for Ulit-BiDet was found to be higher than that of Ulit-BiDet-B by 0.25 (from 0.51 to 0.76). This higher KCC value indicates an enhanced consistency between the classification and localization tasks, achieved through the utilization of the relation-constrained loss. It is worth noting that the incorporation of the relation-constrained loss does not result in an increase in the model parameters. This comparison experiment further affirms the significance of integrating the relation-constrained loss in improving the overall performance of the object detection system.

5) Selection of the Inputs of CAM: In addition, we conducted experiments to assess the effectiveness of different inputs for the CAM module. Specifically, we explored the usage of f_1, f_2, and f_3 to capture edges, along with f_4 to assist in locating object-related edges. The results, as presented in Table IX, demonstrate that the combination of f_1+f_4 achieves the best performance for SAR object detection. This finding can be attributed to the fact that low-level features, such as those captured by f_1, provide abundant edge details. These edge details are particularly crucial for compensating for the reduced representational capacity of the binary network. By incorporating f_4, which helps locate object-related edges, the combined input of f_1+f_4 proves to be the most effective configuration for SAR object detection.

6) Contour Exploration: The visual results of our model with CAM, specifically in terms of contour-related edge extraction, are illustrated in Fig. 8. The figure showcases the visualization results for various scenarios, including small objects, middle-sized objects, large objects, and objects with complex backgrounds, from top to bottom, respectively. Comparing the last column, which represents the extracted contour information f_c, with the third column of $\mathrm{GT}_{\mathrm{contour}}$ (GT contour), it is evident that our method exhibits superior performance in mining object-related contour information. This ability to accurately extract contour details proves to be beneficial for precise object location prediction.

D. Comparison With the SOTA Methods

In order to facilitate more comprehensive comparisons with SOTA detectors, we commence by comparing the performance and scalability of the designed cost-scalable binary backbone with some SOTA neural networks. This comparison is conducted on the ImageNet dataset under identical conditions, and the results are presented in Table X. Subsequently, the proposed detector is evaluated on three distinct datasets, namely, SSDD, AIR-SARShip2.0, and SAR-Ship datasets, using the same conditions. The comparison results are displayed in Table XI. Finally, to enhance the clarity of our optimization techniques' effectiveness, we conduct visual comparisons to evaluate the detection results of the proposed detector on a set of challenging SAR images. These images pose significant difficulty for detection algorithms. The visual comparisons are presented in Figs. 9–11, respectively. By examining these figures, a better understanding of the detector's performance relative to both large-scale and lightweight detectors can be gained.

1) Comparison With SOTA Backbones: The comparison results of the designed backbone with the SOTA neural networks are illustrated in Table X. In order to showcase the achievement of a good balance between performance and complexity in the designed binary backbone networks, we carefully selected a set of SOTA binary networks that exhibit comparable accuracy levels. These networks were used as the basis for our scalable backbone, encompassing a range of sizes from small (S) to large (L). Based on the information provided in Table X, it is evident that the smallest backbone, Backbone-T, demonstrates a slightly lower top-one accuracy of 49.3% compared to XNOR-Net [51] (51.2%). However, it is worth noting that Backbone-T only requires 1/7 of the OPs and 1/4 of the parameters compared to XNOR-Net. In addition, it is important to highlight that Backbone-S and XNOR-Net++, as well as Backbone-M and BiRealNet-34, showcase similar top-one accuracy. However, our proposed backbones achieve this performance while utilizing approximately 1/4 of the OPs and 1/2 of the parameters compared to their counterparts. Finally, it is noteworthy that the largest backbone, Backbone-L, outperforms ReActNet-BiR18 [56] in terms of top-one accuracy, achieving 67.6% compared to

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TABLE XI

COMPARISONS WITH SOTA METHODS ON SAR IMAGES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. "W/A" INDICATES THE BIT WIDTH OF CONVOLUTIONS EXCEPT FOR THE INPUT AND OUTPUT LAYERS

Method	W/A	SSDD			AIR-SARShip2.0			SAR-Ship			Params(M)	OPs(G)
		mAP	Rec	Pre	mAP	Rec	Pre	mAP	Rec	Pre		
SSD512[66]	32/32	94.0	94.4	82.7	75.7	76.8	64.9	94.2	93.1	80.2	24.2	87.7
YOLOv3[50]		95.0	95.3	83.2	68.0	69.9	60.4	94.4	93.3	78.9	61.5	49.6
YOLOv4[36]		96.1	96.4	84.3	**82.1**	**83.6**	**68.3**	93.2	92.5	81.6	64.3	45.3
CenterNet++[84]		95.1	95.2	83.3	73.9	76.4	64.0	94.9	93.8	81.2	20.3	26.6
SSDLite-MNetV2[37]		94.7	93.3	78.7	61.2	66.7	45.6	92.5	91.2	80.9	9.66	2.39
SSDLite-MNetV3[37]		89.8	86.6	73.6	51.3	51.8	56.8	90.0	87.7	73.0	5.71	1.03
RCNNLite-SNetV1[37]	32/32	94.2	93.1	80.7	60.2	61.1	52.8	92.9	92.4	77.4	7.34	2.62
RCNNLite-SNetV2[37]		94.8	94.8	79.3	53.2	56.9	49.6	92.8	92.2	81.2	6.29	1.64
GhostNet[41]		92.1	92.9	83.9	49.0	47.6	57.3	91.7	90.9	82.0	3.61	0.56
Lira-YOLO[27]	32/32	86.3	87.3	70.6	47.8	52.5	47.5	90.9	89.0	83.1	4.30	2.98
NOFA[85]	32/32	89.2	-	-	-	-	-	-	-	-	4.55	-
MEA-Net[86]	32/32	92.8	-	-	-	-	-	-	-	-	0.96	2.80
L-YOLOv4[46]	32/32	95.3	93.9	88.3	68.7	69.3	62.0	95.5	92.6	81.7	43.0	9.30
Quant-Det[37]	8/8	97.0	97.4	**89.1**	77.0	81.2	63.9	**95.7**	**95.3**	**88.7**	3.22	1.40
Ulit-BiDet-T	1/1	91.3	92.2	76.9	68.1	69.8	60.5	84.2	85.7	69.1	**0.56**	**0.10**
Ulit-BiDet-S		93.3	92.5	81.7	70.8	71.9	61.4	86.7	87.5	72.6	**0.74**	**0.14**
Ulit-BiDet-M		95.1	95.1	80.2	71.4	75.7	64.2	89.7	89.8	77.4	**0.96**	**0.19**
Ulit-BiDet-L		**97.2**	**97.6**	**88.9**	77.3	81.1	64.2	95.2	91.8	85.6	**1.27**	**0.22**

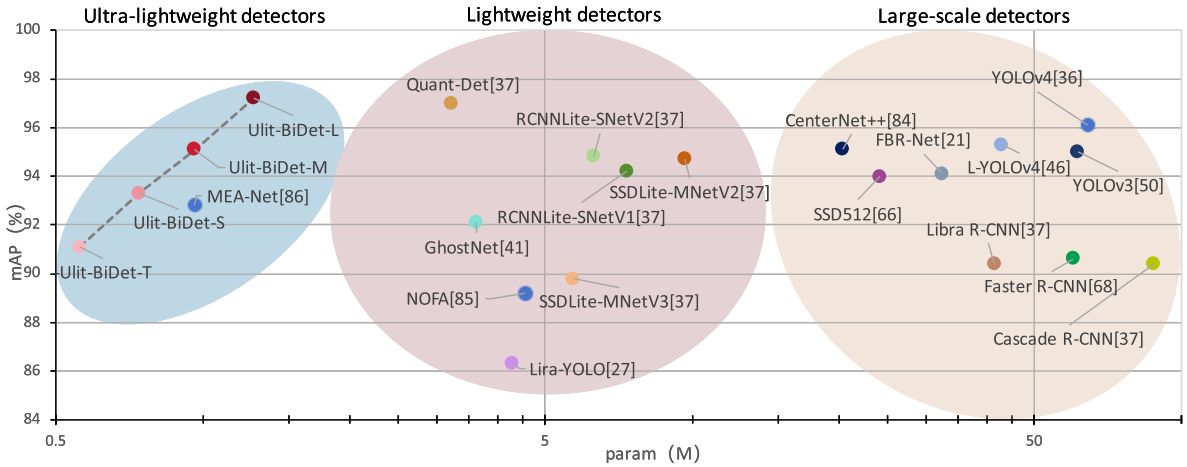


Fig. 9. Comparison of accuracy and model size of the proposed ultralightweight detector with both large-scale detectors and lightweight detectors on the SSDD dataset. To make the concept of "ultralightweight" more clear, we define "lightweight" models as models typically with relatively low computational and storage requirements, typically with parameter quantities defined to be around 5M∼10M [37]. On the other hand, "ultralightweight" models are defined as models with further reduced computational and storage requirements, typically with parameter quantities around 1M [86].

65.9%. In addition, Backbone-L achieves this higher accuracy while utilizing only 1/3 of the OPs and having 0.11M fewer parameters compared to ReActNet-BiR18.

Besides, to further elucidate the achieved tradeoff between the number of parameters, OPs, and top-one accuracy in our backbone, we conducted a comparative analysis between our Backbone-L and comparable real-valued networks, including ResNet18 [81] and ShuffleNetV1 [82]. As shown in Table X, the former serves as the corresponding real-valued network for our backbone baseline, while the latter represents a lightweight network with similar performance to our designed Backbone-L. Specifically, there is only a 3.1% disparity in accuracy between our Backbone-L and ResNet18 (67.6% versus 70.7%) while reducing the large magnitudes of 10.19M params and 4.6G OPs. Moreover, it is noteworthy that Backbone-L's top-one accuracy is comparable to that of the real-value lightweight network ShuffleNetV1 while reducing 2/3 Param and 4/5 OPs.

2) Compared to Large-Scale Detectors: To further validate the superior detectability of our proposed Ulit-BiDet model, we conducted experiments and compared it with commonly used and SOTA large-scale target detection methods on SAR images. The compared methods include SSD512, YOLOv3, YOLOv4, and CenterNet++. In these comparison experiments, we ensured fairness by maintaining consistent parameter settings for each method and using identical datasets and proportions across all experiments. The experimental results are presented in Table XI. In addition, we provide visual comparisons of parameter–accuracy results in Figs. 9–11, respectively. The visualization clearly demonstrates that our proposed lightweight and scalable models can achieve comparable performance to that of large-scale detectors while

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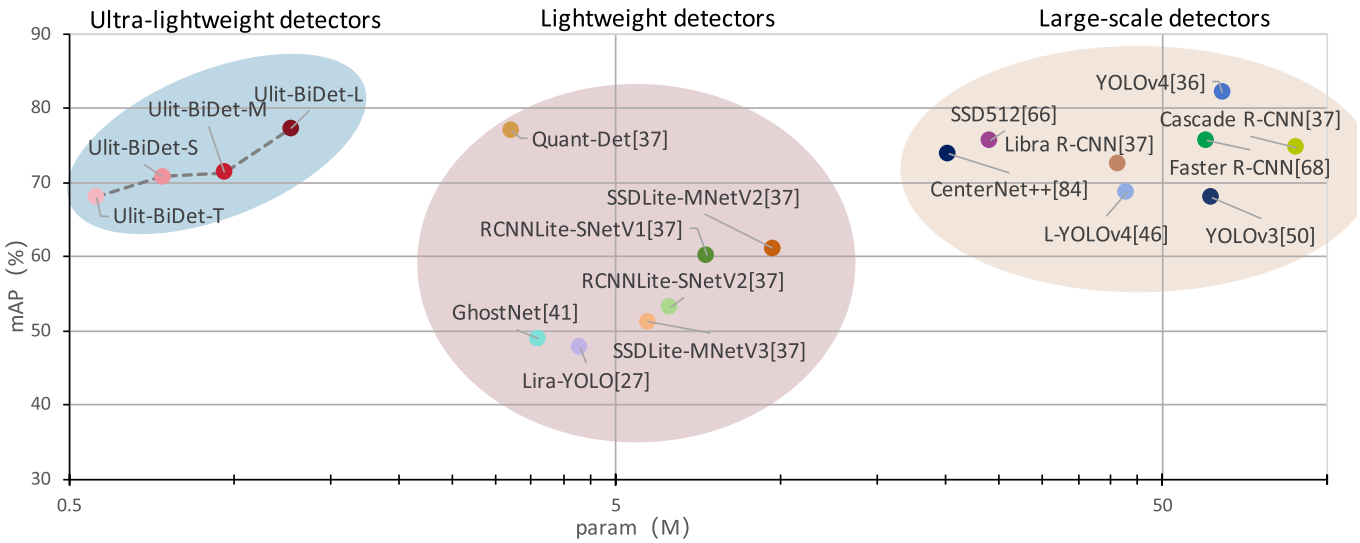


Fig. 10. Comparison of accuracy and model size of the proposed ultralightweight detector with both large-scale detectors and lightweight detectors on the AIR-SARShip2.0 dataset.

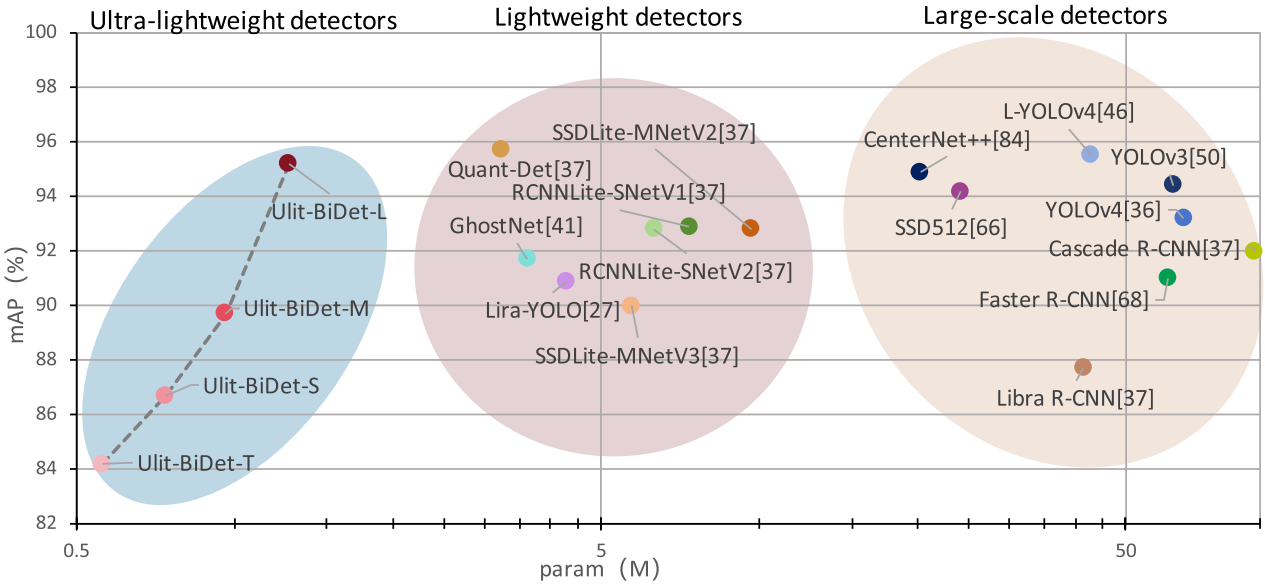


Fig. 11. Comparison of accuracy and model size of the proposed ultralightweight detector with both large-scale detectors and lightweight detectors on the SAR-Ship dataset.

significantly reducing resource requirements. This highlights the efficiency and effectiveness of our approach. In the following, we provide a detailed analysis of the comparison results for further clarification.

As illustrated in Table XI, on the SSDD dataset, we can observe that YOLOv4 has the highest performance among all the large-scale comparison algorithms; however, our Ulit-BiDet-L still achieves a higher mAP than YOLOv4 by 1.1% while only exhibiting 1/60 of the parameter count and 1/200 of the computational cost. The enhancement in performance can be primarily attributed to the incorporation of the nonlocal module in our designed backbone, which improves the contextual representative ability for ship targets. This augmented representation is more conducive to alleviating the interference of land clutter and sea clutter in SAR images, which is consistent with the visualization results in Fig. 13.

Regarding the SAR-Ship dataset, Ulit-BiDet-L achieves a slightly higher mAP compared to CenterNet++, which possesses the highest mAP score among all the large-scale detectors analyzed. This result is noteworthy considering that Ulit-BiDet-L has a significantly lower parameter count of 1/16 and computational costs of 1/120. This clearly emphasizes the substantial advantage of our proposed scheme for practical application deployment.

When considering the AIRSARShip2.0 dataset, it is observed that the mAP of our proposed Ulit-BiDet-L is lower than that of YOLOv4. However, it is important to note that YOLOv4 utilizes a heavy backbone and PANet for feature

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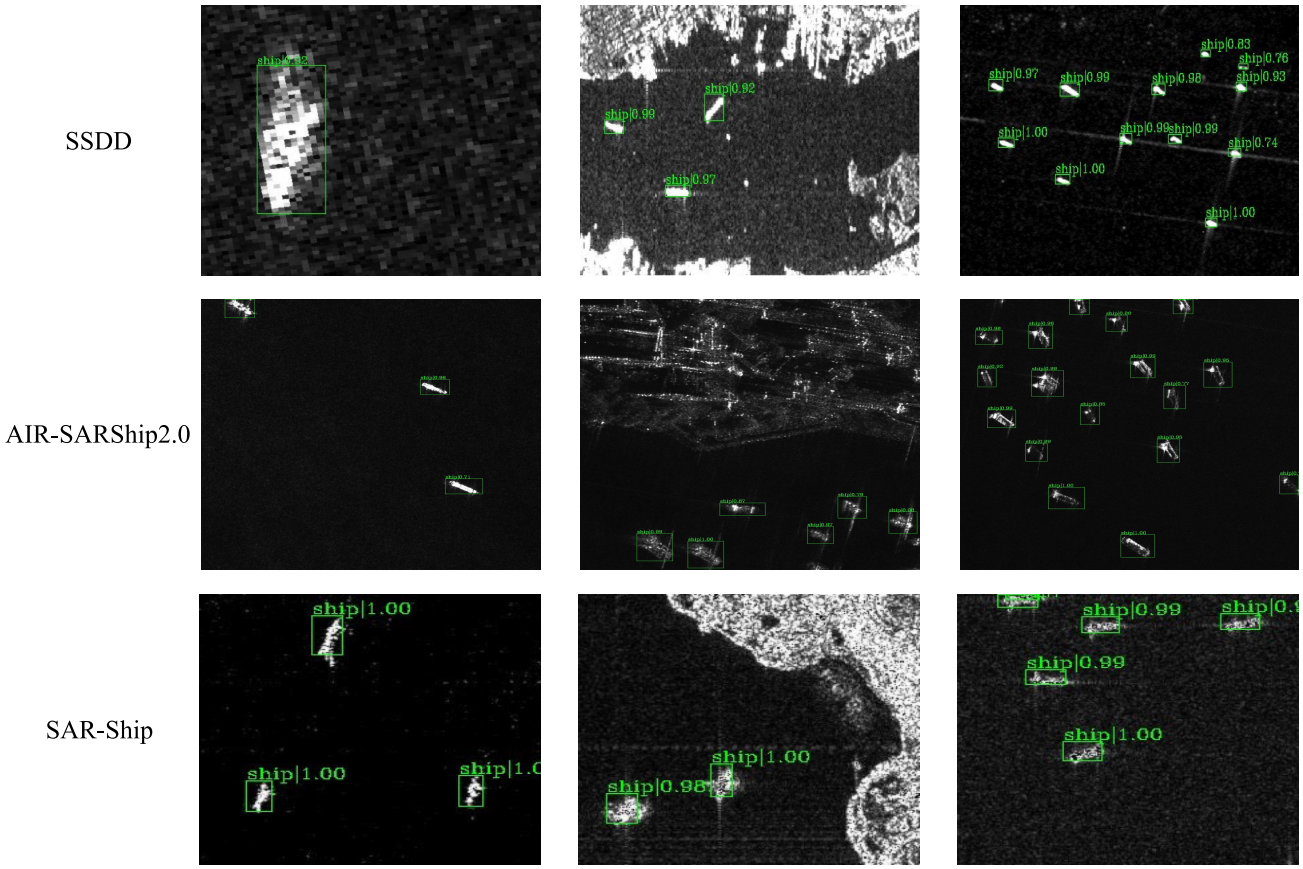


Fig. 12. Visualization of our method on three datasets. The first to third rows show the detection results of the SSDD, AIR-SARShip2.0, and SAR-Ship datasets under different scenarios, including simple background, complex background, and dense target detection, respectively.

extraction, resulting in a much larger parameter count of 61.5 M compared to our Ulit-BiDet-L with only 1.27 M parameters. This contrast highlights the fact that a strong backbone plays a critical role in ship detection, but it contradicts our initial intention of designing a lightweight model.

3) Comparison With Lightweight Detectors: In this section, in order to further substantiate the advantages of the proposed scheme, a series of experiments were conducted to compare its performance and cost with SOTA lightweight target detection methods on SAR images. Two categories of lightweight detectors have been selected: 1) general-purpose networks that have been employed for SAR image detection, namely, SSDLite-MNetV2, SSDLite-MNetV3, RCNNLite-SNetV1, RCNNLite-SNetV2, and GhostNet and 2) detectors that have been specifically optimized for SAR images, including Lira-YOLO, NOFA, MEA-Net, L-YOLOv4, and Quant-Det. Due to the complexity of these methods and the unavailability of open-source code, we have solely referenced the experimental results reported in the respective papers. The collected data are presented in Table XI. Furthermore, we present visual comparisons of the parameter–accuracy results for all the compared schemes in Figs. 9–11, respectively. The visualization reveals that our proposed binary detector achieves comparable performance to that of lightweight detectors while requiring much fewer resources. To provide further clarification, a detailed analysis of the comparison results is presented in the following.

1) In Table XI, the object detection task using lightweight networks is examined. Specifically, two variants of SSD, namely, SSDLite-MNetV2 and SSDLite-MNetV3, are considered computationally efficient options. They make use of MobileNetv2 [87] and MobileNetv3 [88] as their feature extractors. It is claimed that these variants possess eight times fewer parameters than the original SSD while still achieving competitive accuracy. Likewise, the Light-Head RCNN [89] utilizes computationally efficient networks, ShuffleNetV1 [82] and ShuffleNetV2 [90], as its backbone. In this section, these two networks are denoted as RCNNLite-SNetV1 and RCNNLite-SNetV2, respectively.
As depicted in Table XI, our proposed Ulit-BiDet-L consistently outperforms the aforementioned lightweight networks in terms of mAP, recall, and precision across all three datasets. Furthermore, even though GhostNet incorporates feature multiplexing and utilizes only a portion of the channel for convolution to reduce computational requirements, and SSDLite-MNetV3 employs a network search method to achieve a more compact network structure, our proposed Ulit-BiDet-L still demonstrates lower computational costs. Specifically, Ulit-BiDet-L requires 0.56G and 1.03G fewer computations compared to GhostNet and SSDLite-MNetV3, respectively. Moreover, while the performance of

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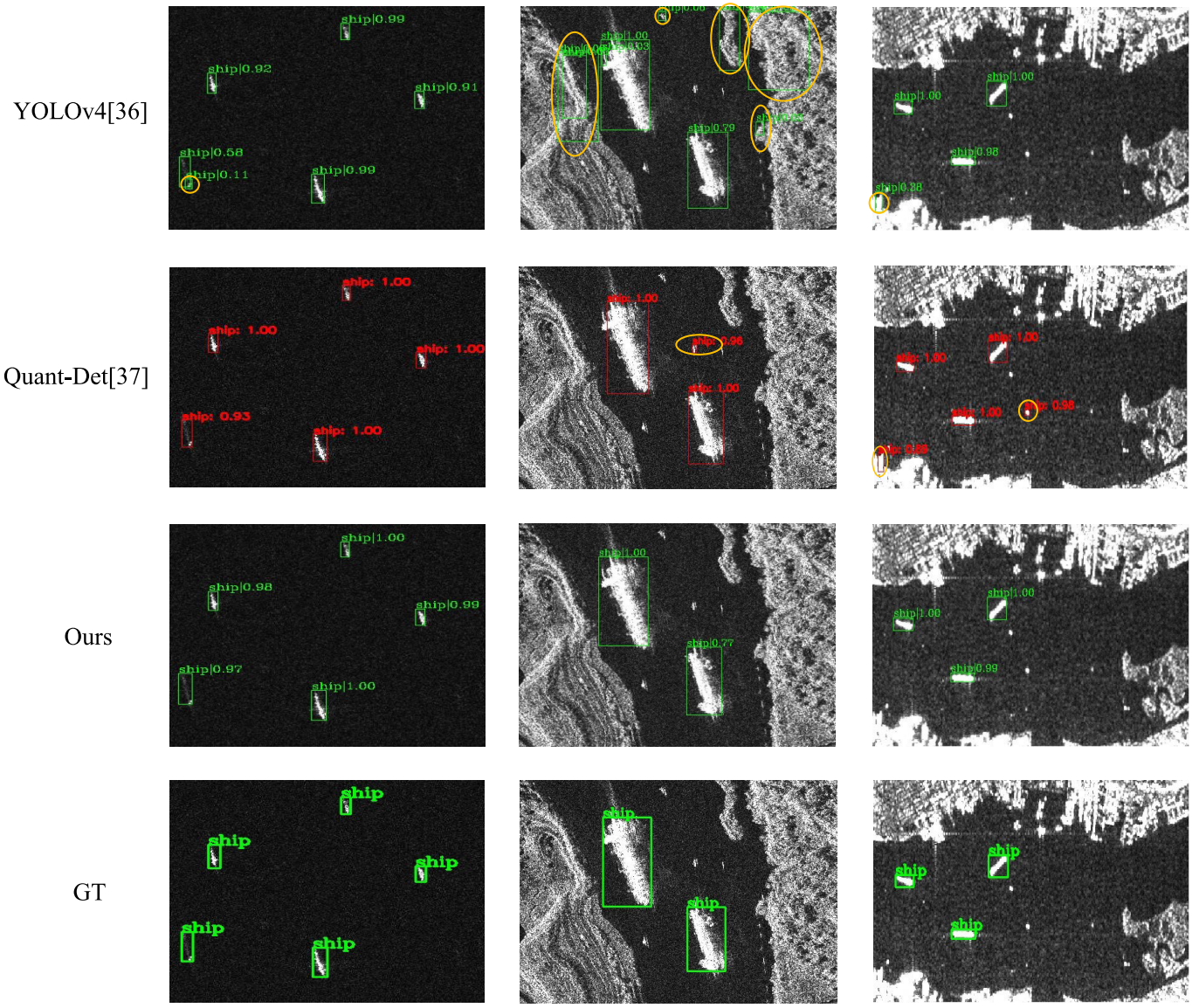


Fig. 13. Comparison of the detection results of different methods on the SSDD dataset. From (top) to (bottom) are the test results of YOLOv4, Quant-Det, our model, and corresponding GT, respectively. The yellow circles represent the false detections.

Ulit-BiDet-T is slightly lower than that of GhostNet on the SAR-Ship dataset, it is worth noting that the params and OPs required by Ulit-BiDet-T are only one-fifth of those needed by GhostNet.

2) In the domain of lightweight SAR object detectors, the latest Quant-Det surpasses Lira-YOLO, NOFA, MEA-Net, and L-YOLOv4, as illustrated in Table XI, thanks to the integration of S-BiFPN and the LTM module. Notably, Quant-Det leverages 8-bit soft quantization to minimize the computational and memory resource requirements. Noteworthy is the fact that our proposed Ulit-BiDet-L, which employs 1-bit quantization, exhibits only a marginal decrease of 0.5% in mAP while utilizing approximately one-third of the parameters and one-seventh of the OPs compared to Quant-Det. Furthermore, Ulit-BiDet-L achieves slightly higher mAP values of 0.2% and 0.3% compared to Quant-Det on the SSDD dataset and the AIR-SARShip2.0 dataset, respectively.

E. Visual Results

In Fig. 12, we present a comprehensive comparison of the detection results using our proposed scheme for different scenarios, including those with simple backgrounds, complex backgrounds, and dense targets from left to right. The first, second, and third rows showcase the detection results on the SSDD, AIR-SARShip2.0, and SAR-Ship datasets, respectively. Remarkably, our method consistently achieves outstanding detection results across diverse scenarios, highlighting the robustness and effectiveness of our approach for various SAR datasets.

Fig. 13 compares the detailed detection results among YOLOv4 [36], Quant-Det [37], and the proposed algorithm in various scenarios on the SSDD dataset. From left to right, the selected scenes include very small objects, large objects, and small objects around complex coasts, respectively. In comparison, YOLOv4 and Quant-Det almost have different degrees of false detection (yellow circles) across all three

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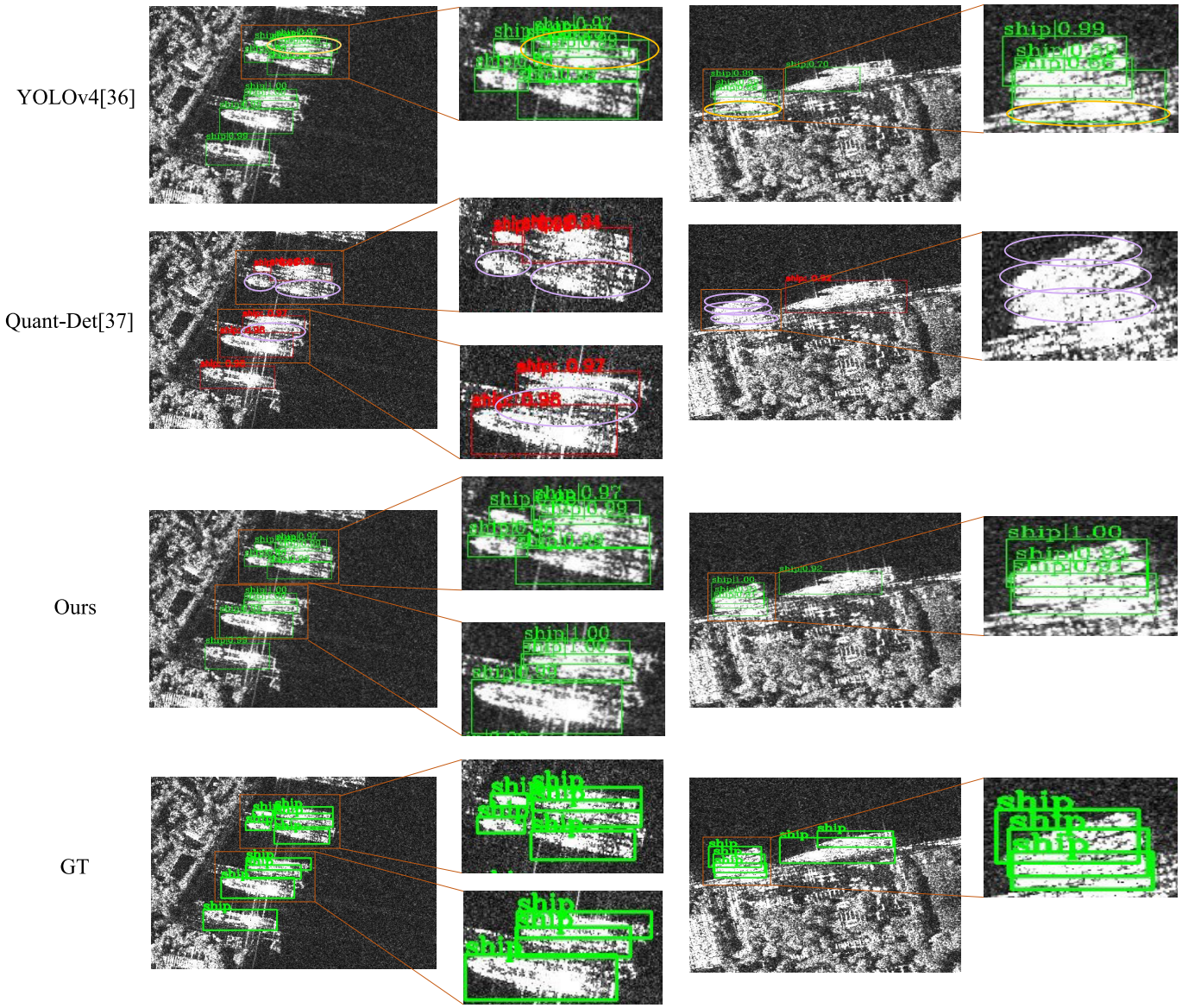


Fig. 14. Comparison of the detection results of different methods on the SSDD dataset. From (top) to (bottom) are the test results of YOLOv4, Quant-Det, our model, and corresponding GT, respectively. The yellow and purple circles represent the false alarms and missing ships, respectively.

scenarios, which is caused by the interference of land clutter and sea clutter, while our method demonstrates superior performance in detecting all targets. The aforementioned analysis of results demonstrates the superior detection capabilities of our proposed method, effectively addressing the challenge of mitigating the interference caused by land clutter and sea clutter across diverse scenes compared with other methods.

Fig. 14 shows the detection results of inshore scenes in the SSDD dataset. Upon observation, it becomes apparent that the other two methods yield inshore false detections (yellow circles), alongside missing several small ships (purple circles). In contrast, our method obtains enhanced performance in detecting ships across varying scales in complex scenes. Meanwhile, for some ships, our method achieves more accurate bounding boxes. As denoted by the yellow and purple circles, the detection results of YOLOv4 and Quant-Det cannot cover the intact object accurately. This inadequacy can be attributed to the vulnerability of the semantic representations of local areas close to the ship target contour, which are prone to interference from neighboring regions with similar scattering power. In contrast, our proposed model effectively mitigates such interference, resulting in improved detection performance.

Despite the various experimental results have validated the efficiency of our lightweight target detection network, the method still has a lot of room for exploration in different settings, i.e., considering the rotated bounding boxes or using the original single-look complex (SLC) images or even raw SAR data to directly target sensing when designing the method. In future studies, we will incorporate the original SLC images or even raw SAR data to extract the special SAR characteristics for ship target sensing [13], [14]. Besides, we will consider extending our ultralightweight ship detection model to SAR ship images for rotated detection [91], [92].

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V. Conclusion

In this article, we build an efficient and ultralightweight binarized ship detector in SAR images, denoted as Ulit-BiDet, which uses binary quantization techniques to effectively reduce the number of parameters and computational effort of the model. First, the designed performance and cost-scalable binary backbone satisfy various resources and computational power of practical devices. In addition, to better fit the characteristics of SAR image ship data, the nonlocal module in the designed backbone is introduced to effectively mitigate the interference caused by land clutter and sea clutter. Then, we propose a contour-information enhancement module, including the CAM and CFM, to leverage object-related contour semantics, which benefits the precise localization of ship targets. Finally, we incorporate relation-constrained loss to refine the alignment between the target classification and its corresponding location. Experimental results show that our method achieves a performance comparable to SOTA detectors while using fewer parameters and computational resources. In future work, we will focus on implementing Ulit-BiDet in dedicated hardware accelerators, such as embedded field programmable gate array (FPGA) devices, to further evaluate the performance gain in execution speed.

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Han Pu received the master's degree from Fujian Normal University, Fuzhou, China, in 2020. She is currently pursuing the Ph.D. degree with the Institute of Information Science, Beijing Jiaotong University, Beijing, China.

Her research interests include neural network compression, binary representation, and computer vision.



Zhengwen Zhu received the B.S. degree in network engineering from Henan Polytechnic University, Jiaozuo, Henan, China, in 2021. He is currently pursuing the M.S. degree with the Institute of Information Science, Beijing Jiaotong University, Beijing, China.

His research interests include object detection and lightweight models.



Qi Hu received the B.S. and M.S. degrees from the College of Electronic and Information Engineering, Hebei University, Baoding, China, in 2016 and 2019, respectively. She is currently pursuing the Ph.D. degree with the Institute of Information Science, Beijing Jiaotong University, Beijing, China.

Her research interests include computer vision, image processing, and target detection.



Dong Wang (Member, IEEE) received the M.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 2006 and 2010, respectively.

He was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of California at Davis, Davis, CA, USA, from 2018 to 2019. He is currently a Professor with the School of Computer and Information Technology, Beijing Jiaotong University. His research interests include reconfigurable computing, high-performance computing architectures for embedded applications, and computer vision.

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